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Final Report

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Abstract

Floods and landslides are natural hazards that affect Sri Lanka especially during times of heavy and long lasting rain. Traditional methods for warning about disasters use observations from the environment rainfall levels, past disaster records and expert judgment. This project builds an AI based prototype to help assess flood and landslide risks by combining three intelligence methods: Machine Learning, Fuzzy Logic and Rule Based Reasoning.

Data from climate and disaster events between 2007 and 2020 were gathered and combined with terrain information taken from elevation data. The final dataset includes 122,736 observations, one for each district on a given day covering 24 districts across Sri Lanka. Eight environmental factors were used in the model: rainfall over 24 hours rainfall over 72 hours rainfall over 7 days average temperature, dew point temperature, relative humidity, mean elevation and mean slope.

Two machine learning models Decision Tree and Random Forest were tested for predicting both floods and landslides. After evaluation the Random Forest model was chosen for the system because it performed better overall. Threshold values were determined using validation data with 0.70 set as the threshold for flood prediction and 0.80 for landslide prediction.

The final intelligent system brings together the probability outputs from the Random Forest model with fuzzy logic risk assessment and rule-based reasoning based on rainfall to produce a risk level: LOW, MEDIUM or HIGH. A user-friendly interface built using Streamlit allows users to input weather and environmental conditions, choose a district and get updated risk information for floods, landslides and overall disaster risk.

This prototype shows how different AI techniques can work together to support decisions related to disaster risk. However due to severe class imbalance,limited historical disaster records data aggregated at the district level and constraints in the available historical data the system should be seen as an academic decision-support tool rather than a replacement, for official disaster warnings.

1. Introduction

Sri Lanka frequently faces floods and landslides especially when the monsoon rains are heavy. These calamities can damage homes, roads, farms and transport routes. They can also push people out of their homes cause injuries and result in deaths.

Flood risk and landslide risk depend on environmental conditions that work together. How hard the rain falls and how much rain has collected are factors. The shape of the land such as how high it's how steep it is can also affect how likely a flood or landslide is.

Artificial Intelligence offers methods that can spot patterns in data and help make decisions when things are uncertain. Therefore this project looks at whether old climate data, land data and past disaster records can be joined to estimate flood risk and landslide risk.

The system was built as an AI solution that uses three main AI ideas. First Machine Learning with Random Forest classification. Second Fuzzy Logic to manage uncertainty and overlapping conditions. Third Rule-Based Reasoning that applies rules triggered by rainfall.

These parts are combined into one prototype. It gives assessments, for flood and landslide and it also shows an overall risk level.

2. Problem Background

Floods and landslides happen often in Sri Lanka. When there is a lot of rain water runs quickly over the land. Causes floods. When it rains for a time the hills become unstable and landslides can happen.

One big problem is that disasters happen because of things at the same time. For example just having rain does not always mean a landslide will happen. The shape of the land how much rain has already. The weather can all play a role.

Another problem is that it is not always clear what is happening. Environmental conditions are not always easy to put into groups. For example there is no moment when rain changes from "medium" to "high." Because of this fuzzy logic helps to show how different conditions mix together.

Because of these issues the project started as a way to sort things and then became a system that uses past patterns, fuzzy thinking and set rules, about rain.

3. Problem Statement

There is a need for an AI-based decision-support system that can analyze environmental and terrain conditions and estimate flood and landslide risk. A rule-based approach may fail to capture complex patterns in historical data. A machine-learning-based system can be hard to interpret and may give uncertain results especially when disaster observations are limited.

The problem this project seeks to address is: How can Machine Learning, Fuzzy Logic and Rule-Based Reasoning be combined to analyze environmental and disaster data and provide clear flood and landslide risk levels for districts, in Sri Lanka?

The system is developed as a decision-support prototype. It is not intended to replace warnings issued by authorities. Instead it aims to support decision-making with interpretable risk assessments.

4. Aim and Objectives

4.1. Aim

The aim of this project is to design and implement an AI-based system capable of assessing flood and landslide risk using historical climate and terrain information.

4.2. Objectives

The objectives are:

- To collect and preprocess historical climate and disaster data for Sri Lanka.
- To integrate rainfall, temperature, humidity and terrain characteristics into a unified dataset.
- To derive terrain-related features such as mean elevation and mean slope.
- To develop separate Machine Learning classifiers for flood and landslide occurrence.
- To compare Decision Tree and Random Forest classification techniques.
- To select appropriate probability thresholds using validation data.
- To implement fuzzy logic for handling uncertainty in rainfall, slope and Machine Learning probabilities.
- To implement rainfall-based rule reasoning as an additional decision layer.
- To integrate the three AI components into a single hybrid risk-assessment system.

- To classify flood and landslide risk into understandable LOW, MEDIUM and HIGH levels.
- To develop a simple Streamlit interface through which users can enter environmental conditions and obtain predictions.
- To evaluate the system using unseen historical test data and identify its limitations.

5. Selected AI Techniques

5.1 Machine Learning Classification

Two classification algorithms were investigated:

- Decision Tree
- Random Forest

Separate classifiers were developed for **Flood occurrence and Landslide occurrence**

The Machine Learning component produces probabilities indicating the estimated likelihood of each disaster based on the supplied environmental conditions.

Random Forest was selected for the final system after comparison with the Decision Tree models.

5.2 Fuzzy Logic

Fuzzy logic was incorporated to handle uncertainty in environmental conditions.

Three variables are considered by the fuzzy component:

- Rainfall
- Terrain slope
- Machine Learning probability

Each variable is represented using overlapping fuzzy membership functions.

For example, rainfall is represented using:

- LOW
- MEDIUM
- HIGH

Similarly, slope and Machine Learning probability contain LOW, MEDIUM and HIGH membership categories.

The implemented rainfall membership ranges include overlapping low, medium and high functions, while slope and model probability are treated similarly. app.py

The fuzzy system produces a numerical risk score using representative values:

- LOW = 20
- MEDIUM = 55
- HIGH = 90

The resulting score is converted into:

- Score below 40 → LOW
- Score from 40 to below 70 → MEDIUM
- Score of 70 or above → HIGH

5.3 Rule-Based Reasoning

A rule-based rainfall warning layer was added to provide additional reasoning independent of the Machine Learning prediction.

The implemented rainfall rules are:

24-hour Rainfall	**Rule Status**
< 75 mm	NORMAL
75–99.99 mm	LOW-WATCH
100–149.99 mm	MEDIUM
≥ 150 mm	HIGH

The implementation uses these thresholds directly in the rule-based function. app.py

If the rainfall rule indicates HIGH, the landslide risk is raised to HIGH. If it indicates MEDIUM while the calculated landslide risk is LOW, the system raises the landslide risk to MEDIUM.

6. Justification of AI Techniques

6.1 Random Forest

Flood and landslide prediction can be considered as binary classification problems because the historical data shows whether a flood or landslide occurred on a particular date and in a particular district.

Random Forest was selected because it can handle complex and nonlinear relationships between different environmental factors. It also combines several decision trees, which generally makes the model more stable than using only one decision tree.

During the model comparison, Random Forest gave better overall results than the individual Decision Tree. This was especially noticeable when comparing the models using ranking-based measures such as ROC-AUC.

6.2 Fuzzy Logic

Environmental conditions do not always have clear boundaries. For example, rainfall of 99 mm and 101 mm are almost the same in terms of the actual environmental condition. However, if a fixed threshold is used, these two values could be placed into two different categories.

Fuzzy logic helps to deal with this type of uncertainty. Instead of placing a value completely into one category, it allows a value to have different levels of membership.

In this project, fuzzy logic was used to represent conditions such as:

- Low rainfall
- Moderate rainfall
- High rainfall
- Gentle slope
- Steep slope
- Low model probability
- High model probability

This makes the final risk assessment more flexible than using only fixed thresholds.

6.3 Rule-Based Reasoning

The Machine Learning models learn patterns from historical data. However, the disaster data is highly imbalanced, with many more non-disaster observations than actual disaster observations.

Because of this imbalance, it would not be appropriate to depend only on the Machine Learning predictions. A rule-based layer was therefore added to provide an additional decision mechanism based on rainfall conditions.

The final system combines three different approaches:

- **Machine Learning** – identifies patterns from historical data.
- **Fuzzy Logic** – handles uncertainty in environmental conditions.
- **Rule-Based Reasoning** – applies simple rainfall-based decision rules.

7. System Design

7.1 System Architecture

The system follows a step-by-step process starting from historical data and ending with a final disaster risk level.

The overall workflow is:

Historical Climate Data + Historical Disaster Data + Terrain Data

↓

Data Cleaning and Pre-processing

↓

Feature Engineering

↓

Flood Random Forest + Landslide Random Forest

↓

ML Probabilities

↓

Fuzzy Logic Risk Assessment

↓

Rainfall Rule-Based Reasoning

↓

Final Flood Risk + Landslide Risk

↓

Overall Risk

↓

Streamlit User Interface

This architecture allows the system to use Machine Learning predictions together with fuzzy logic and predefined rainfall rules.

8. Implementation Details

8.1 Data Preparation

Historical climate and disaster data from 2007 to 2020 were used for the project.

Before combining the datasets, the dates and district names were standardized. The disaster records were then converted into daily district-level binary values for:

- Flood
- Landslide
- Any Disaster

The climate and disaster datasets were merged using **Date** and **District**.

8.2 Missing Rainfall Investigation

During the data cleaning process, 5,114 rows were found to have missing rainfall information.

After investigating these missing values, it was found that all 5,114 observations belonged to Colombo District and covered the study period.

Among these missing records, there were:

- 38 flood district-days
- 6 landslide district-days
- 44 total disaster district-days

Instead of creating artificial rainfall values through imputation, Colombo District was removed from the modelling dataset. This decision was made because rainfall was one of the main environmental variables used by the model, and rainfall information was completely unavailable for Colombo during the study period.

After removing Colombo, the final modelling dataset contained **24 districts**.

8.3 Terrain Data

Terrain information was included because geographical conditions can influence both flood and landslide risk.

A NASADEM Digital Elevation Model (DEM) was used to obtain elevation and slope information.

The source raster had the following properties:

- CRS: EPSG:4326
- Width: 8,640 pixels
- Height: 14,400 pixels
- Approximate resolution: 0.00027778°
- Data type: int16
- NoData value: -32768

The DEM was re-projected to **EPSG : 32644** so that terrain calculations could be performed using metres. The resulting resolution was approximately **30.69 m**.

Slope was calculated using the elevation gradient. The final national slope raster had approximately:

- Minimum slope: 0°
- Maximum slope: 76.66°
- Mean slope: 3.65°

The implementation reads the NASADEM raster, re-projects it and calculates the slope from the elevation data.

District boundaries were then used to calculate terrain statistics for each district.

The two terrain features used in the final model were:

- **Mean Elevation (m)**
- **Mean Slope (deg)**

8.4 Final Dataset

After combining the climate, disaster and terrain information, the final dataset contained:

- **122,736 observations**
- **24 districts**

The model uses eight input features:

Feature	Description
Rainfall_24h	Rainfall accumulated during the previous 24 hours
Rainfall_72h	Rainfall accumulated during the previous 72 hours
Rainfall_7day	Rainfall accumulated during the previous 7 days

Mean_Temp_C	Mean temperature
Dew_Temp_C	Dew-point temperature
Relative_Humidity	Relative humidity
Mean_Elevation_m	Mean elevation of the district
Mean_Slope_deg	Mean slope of the district

The two target variables are:

- **Flood**
- **Landslide**

8.5 Train, Validation and Test Split

A chronological data split was used instead of randomly mixing the observations.

This approach was selected because it provides a more realistic evaluation. The model is trained using earlier observations and then tested using later observations that were not used during training.

Dataset	Rows	Flood Events	Landslide Events
Training	96,432	713	471
Validation	8,760	91	12
Testing	17,544	106	33

This means that the final test results represent the model's performance on later unseen observations.

8.6 Model Development

Both Decision Tree and Random Forest classifiers were developed separately for flood and landslide prediction.

Since disaster events were much less common than non-disaster observations, accuracy alone was not considered sufficient for evaluating the models.

The following evaluation measures were used:

- Precision
- Recall
- F1-score
- Confusion Matrix
- PR-AUC
- ROC-AUC

These metrics provide a better understanding of how well the models identify the relatively small number of actual disaster events.

8.7 Threshold Selection

The default classification threshold of 0.50 was not used automatically.

Different probability thresholds were tested using the validation dataset to find a more suitable threshold for each model.

For the flood model, the best validation F1-score was obtained at:

Flood threshold = 0.70

For the landslide model:

Landslide threshold = 0.80

These thresholds were fixed before testing the models on the unseen test dataset.

Therefore, the final model uses:

- **0.70 for flood prediction**
- **0.80 for landslide prediction**

This approach prevents the test dataset from being used to select the final thresholds.

8.8 Hybrid Decision Process

When a new observation is entered into the system, the following process takes place.

First, the eight environmental features are given to both Random Forest models.

The flood model produces a flood probability, while the landslide model produces a landslide probability.

Next, rainfall, slope and the Machine Learning probabilities are processed using fuzzy membership functions.

A fuzzy risk score is then calculated and converted into one of three categories:

- LOW
- MEDIUM
- HIGH

The selected Machine Learning thresholds are then applied to the model probabilities.

After that, the rainfall-based rule is checked.

Finally, the flood and landslide results are combined. The higher risk category between the two becomes the overall disaster risk.

Therefore, the application provides:

- Flood probability
- Flood risk level
- Flood fuzzy score
- Landslide probability
- Landslide risk level
- Landslide fuzzy score
- Rule status
- Overall risk

8.9 User Interface

A graphical prototype was developed using Streamlit.

The user can select a district and enter the required environmental information.

The available inputs include:

- District
- 24-hour rainfall
- 72-hour rainfall
- 7-day rainfall
- Mean temperature
- Dew-point temperature
- Relative humidity

The system also displays the selected district's mean elevation and mean slope.

After the user runs the prediction, the application displays:

- Flood risk
- Flood ML probability
- Flood fuzzy risk score
- Landslide risk
- Landslide ML probability
- Landslide fuzzy risk score
- Overall disaster risk
- Rule-based rainfall status

The GUI therefore provides both the input values and the final risk results in one place.

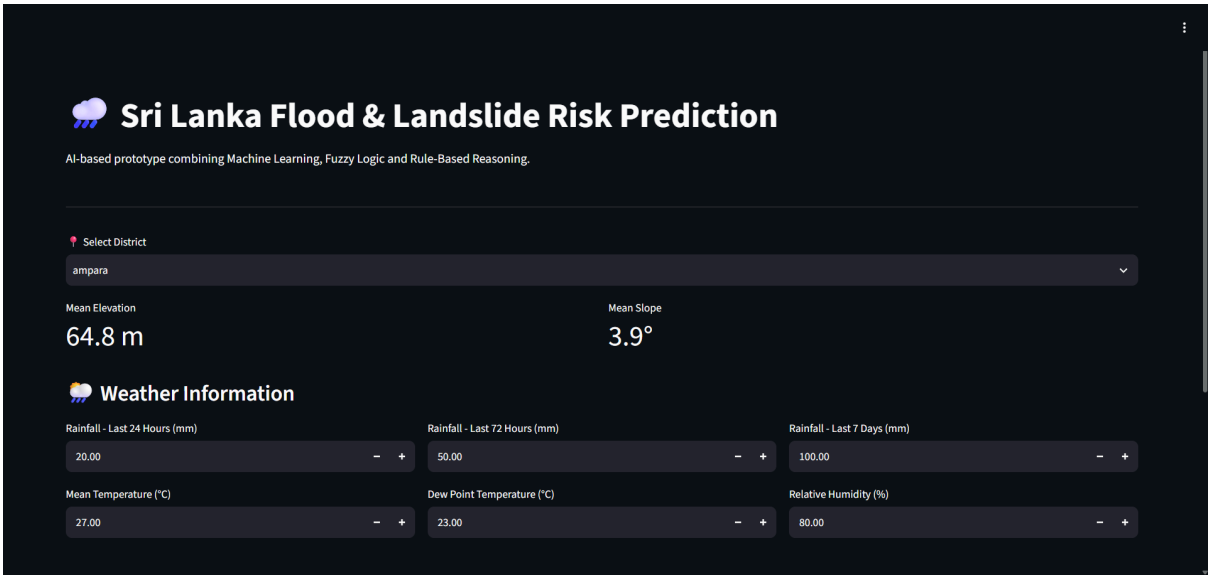


Figure 1: Main Dashboard of the AI-Based Flood and Landslide Risk Prediction System

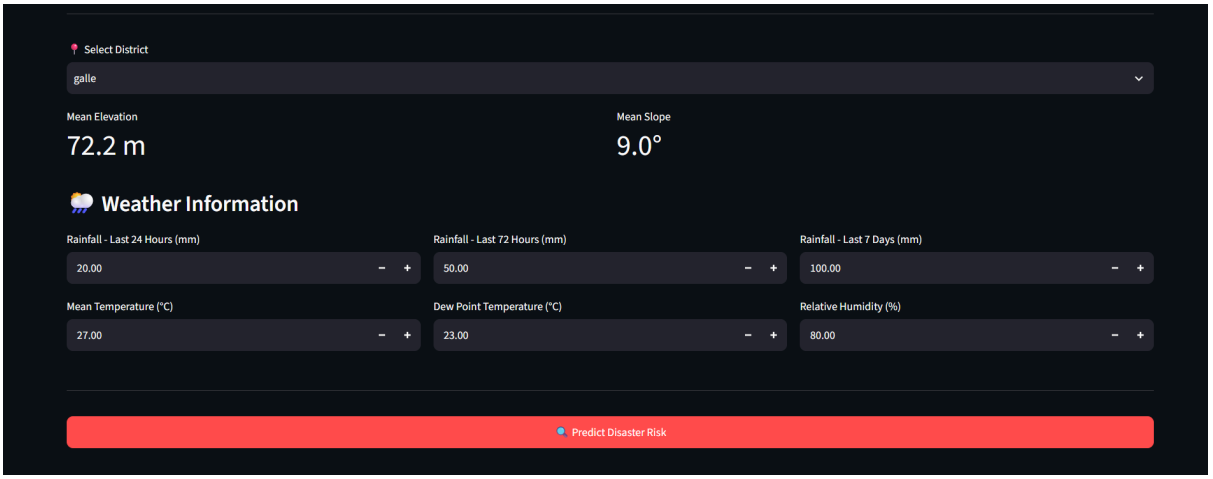


Figure 2: Streamlit User Interface Showing District Selection and Weather Data Inputs

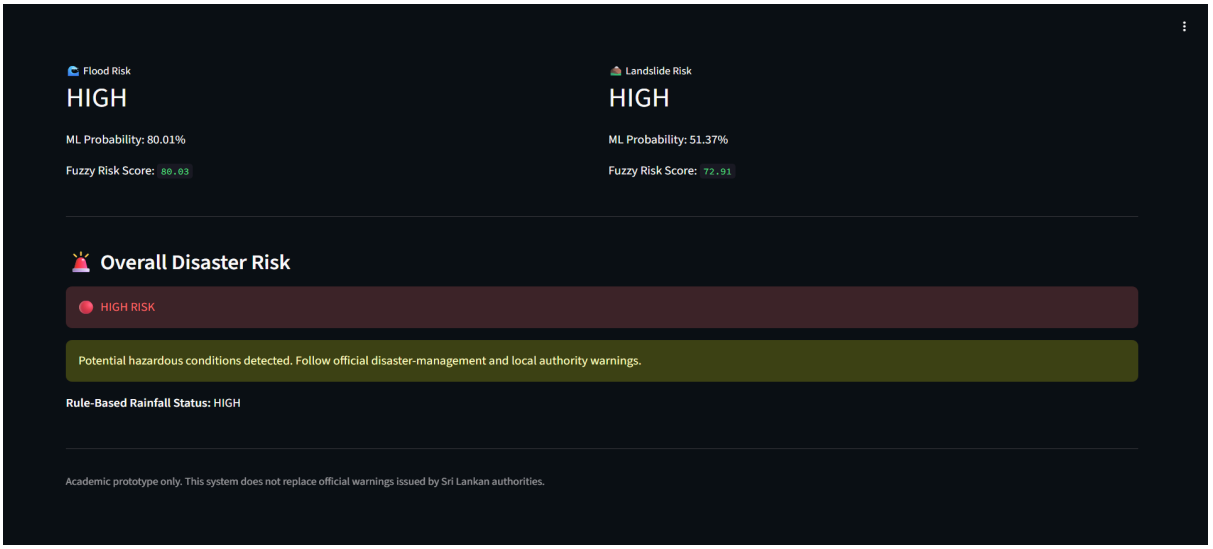


Figure 3: High Flood and Landslide Risk Prediction Results with Overall High Disaster Risk

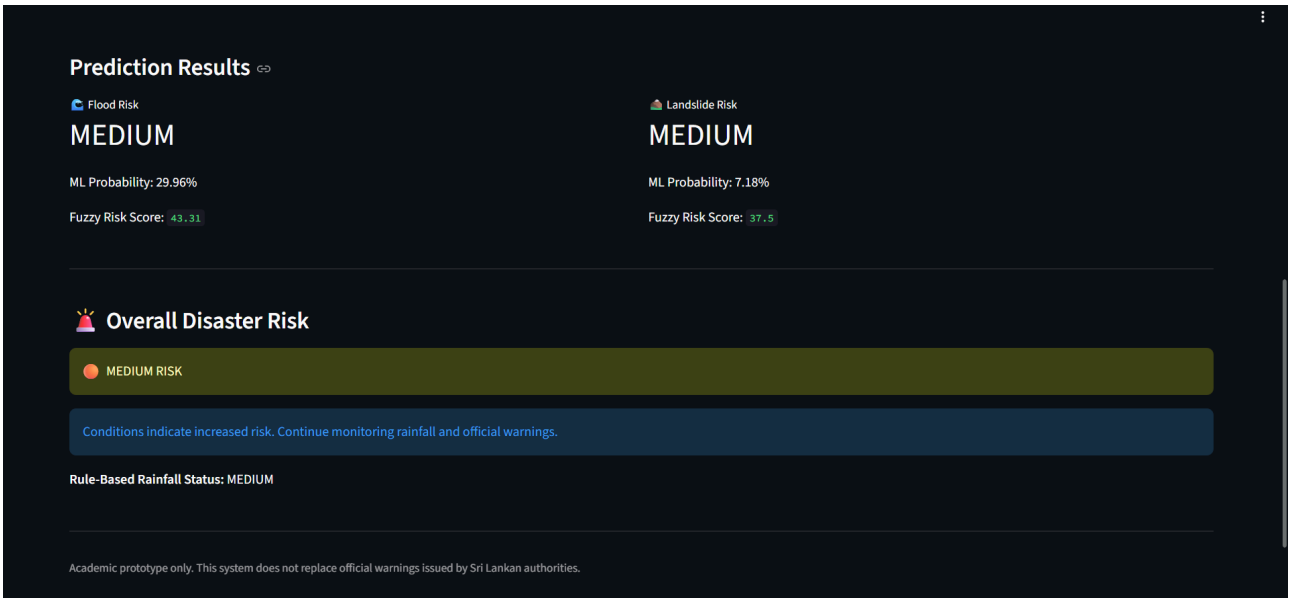


Figure 4: Medium Flood and Landslide Risk Prediction Results with Overall Medium Disaster Risk

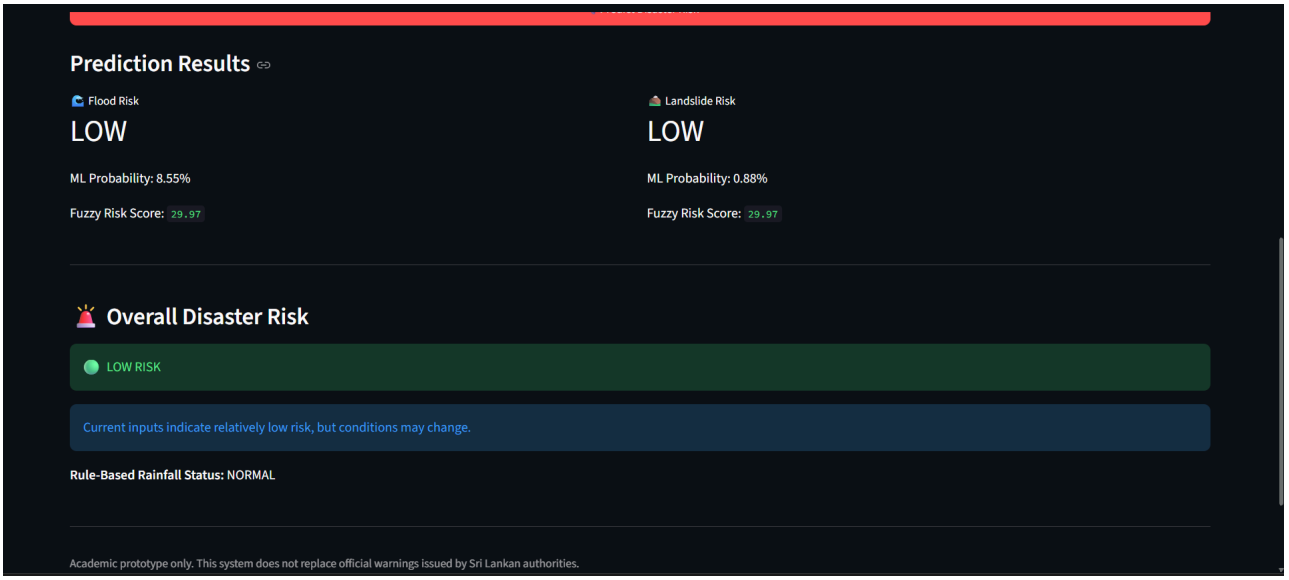


Figure 5: Low Flood and Landslide Risk Prediction Results with Overall Low Disaster Risk

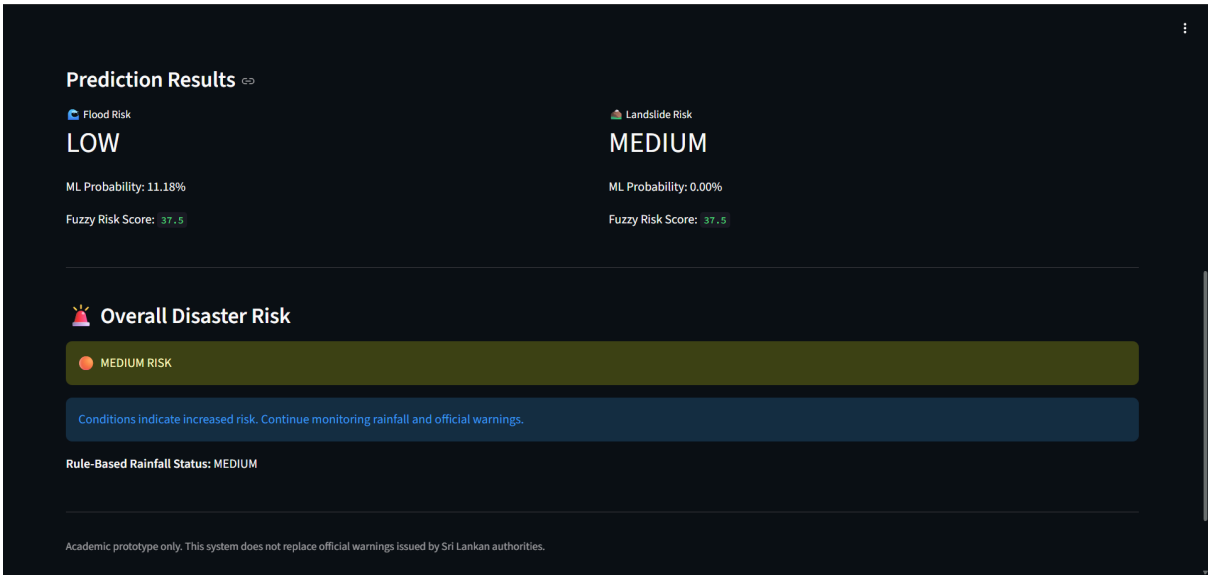


Figure 6: Low Flood Risk and Medium Landslide Risk Prediction Results

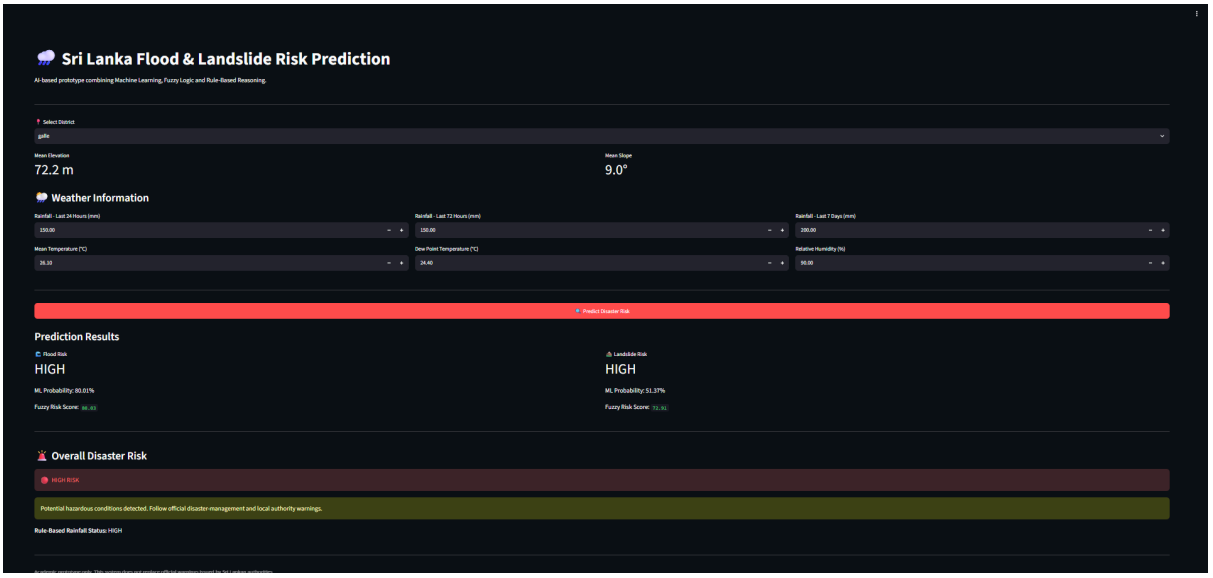


Figure 7: Complete System Demonstration Showing Weather Inputs and High Disaster Risk Prediction Results

Updated System Features-: The Streamlit application was enhanced with additional input validation and usability features. The system checks the consistency of cumulative 24-hour, 72-hour, and 7-day rainfall measurements and displays warnings for potentially inconsistent temperature and dew-point values. A reset button was also implemented to restore weather inputs to their default values, while prediction timestamps are displayed in Sri Lanka Standard Time. Additionally, error-handling mechanisms provide informative messages when required model or dataset files cannot be loaded.

These improvements enhance input data quality, usability, and system robustness. However, validation warnings do not prevent predictions from proceeding.

The implementation and related code changes are available in the project's GitHub repository.

9. Testing and Results

9.1 Flood Decision Tree – Validation

The Flood Decision Tree produced the following results:

- Accuracy: **81.60%**
- Disaster precision: **3.26%**
- Disaster recall: **58.24%**
- Disaster F1-score: **6.17%**

The model was able to identify a reasonable number of flood events. However, it also produced a large number of false-positive predictions.

9.2 Flood Random Forest – Validation

The Flood Random Forest produced:

- Accuracy: **94.25%**
- Flood precision: **7.42%**
- Flood recall: **39.56%**
- Flood F1-score: **12.50%**
- PR-AUC: **0.1215**
- ROC-AUC: **0.8411**

After testing different probability thresholds, **0.70** was selected as the best validation threshold based on F1-score.

At this threshold, the results were:

- Precision: **18.00%**
- Recall: **19.78%**
- F1-score: **18.85%**

9.3 Landslide Decision Tree – Validation

The Landslide Decision Tree produced:

- Accuracy: **86.34%**
- Landslide precision: **0.83%**
- Landslide recall: **83.33%**
- Landslide F1-score: **1.64%**

Although the recall was high, the model generated a very large number of false-positive warnings.

9.4 Landslide Random Forest – Validation

The Landslide Random Forest produced:

- Accuracy: **94.82%**
- Landslide precision: **1.75%**
- Landslide recall: **66.67%**
- Landslide F1-score: **3.40%**
- PR-AUC: **0.0438**
- ROC-AUC: **0.9000**

The threshold analysis selected **0.80** based on the validation F1-score.

At this threshold:

- Precision: **7.32%**
- Recall: **25.00%**
- F1-score: **11.32%**

9.5 Final Flood Model – Unseen Test Data

The final Flood Random Forest was evaluated on the unseen test dataset using the fixed threshold of **0.70**.

The confusion matrix was:

	Predicted No Flood	Predicted Flood
ActualNo Flood	17,256	182
Actual Flood	79	27

The final results were:

- Accuracy: **98.51%**
- Flood precision: **12.92%**
- Flood recall: **25.47%**
- Flood F1-score: **17.14%**
- PR-AUC: **0.0843**
- ROC-AUC: **0.9150**

9.6 Final Landslide Model – Unseen Test Data

The final Landslide Random Forest was evaluated using the fixed threshold of **0.80**.

The confusion matrix was:

	Predicted No Landslide	Predicted Landslide
Actual No Landslide	17,453	58
Actual Landslide	24	9

The final results were:

- Accuracy: **99.53%**
- Landslide precision: **13.43%**
- Landslide recall: **27.27%**
- Landslide F1-score: **18.00%**
- PR-AUC: **0.0752**

- ROC-AUC: **0.9287**

9.7 Interpretation of Results

The high accuracy values should not be taken to mean that the models can correctly predict almost every disaster.

The main reason is that the dataset is highly imbalanced. For example, the test dataset contains only **106 flood events out of 17,544 observations** and only **33 landslide events**.

Because most observations belong to the non-disaster class, a model can achieve high accuracy by correctly predicting the majority class.

Therefore, precision, recall, F1-score, PR-AUC and ROC-AUC are more useful when evaluating these models.

The Flood Random Forest achieved a test ROC-AUC of **0.915**, while the Landslide Random Forest achieved **0.9287**. These results show that the models have useful ability to rank observations from lower risk to higher risk.

However, the relatively low precision and recall values for the positive class show that the models are not reliable enough to be used independently as an operational disaster warning system.

For this reason, the project combines the Machine Learning predictions with fuzzy logic and rainfall-based rules.

9.8 Prototype Testing

Historical disaster observations from the test dataset were also passed through the complete hybrid system.

For example, several historical flood observations from Ampara during October to December 2019 produced relatively high flood probabilities.

Some of the probabilities were approximately:

- 0.7713

- 0.7336
- 0.7932
- 0.8727
- 0.7996
- 0.8472

After applying the updated hybrid risk logic, many of these historical flood observations were classified as **HIGH overall risk**.

This shows that the system can combine Machine Learning probabilities and environmental conditions and convert them into understandable risk categories.

However, some actual disaster observations were still classified as LOW or MEDIUM. This indicates that the current system still needs further improvement.

10. Limitations and Discussion

There are several limitations in the current system.

Class Imbalance

Disaster observations make up only a small part of the dataset. This makes the classification problem difficult and explains why the models can achieve high accuracy while still having relatively low precision and recall for actual disaster events.

Limited Landslide Examples

There were only 12 landslide events in the validation dataset and 33 in the test dataset. Because the number of positive examples is small, the landslide performance metrics can change significantly with only a small number of predictions.

Colombo Exclusion

Colombo District was removed because all 5,114 rainfall observations for the district were missing. Therefore, the current model covers 24 districts instead of all 25 districts in Sri Lanka.

District-Level Terrain Data

Elevation and slope were represented using district-level average values. However, terrain conditions can vary significantly within a district. Therefore, the current terrain information cannot accurately represent individual hillsides or specific flood-prone locations.

Historical Data Limitations

Historical disaster records may not contain every disaster event. The completeness of the records can also vary between different areas and time periods.

No Real-Time Data

The current Streamlit prototype requires users to enter weather conditions manually. It is not connected to a live weather or meteorological data source.

Rainfall Rules

The rainfall rules used in this project are simplified rules developed for the academic prototype. They should not be considered a replacement for the official methodologies used by organizations such as NBRO or DMC.

Fuzzy Membership Functions

The fuzzy membership boundaries were designed specifically for this prototype. Further testing and validation with environmental or disaster-management experts would be required.

Model Generalization

The models were trained using data from 2007–2020. Weather patterns, land use and environmental conditions can change over time. Therefore, the models may need to be retrained using newer data.

Prediction vs Official Warning

The system only estimates risk using a limited number of environmental variables. An official disaster warning requires additional data, expert analysis and professional decision-making.

During development, several technical challenges were also encountered. These included memory limitations when processing high-resolution terrain data and changes in the availability of the online district-boundary service. To overcome these issues, the workflow was adjusted and previously generated terrain outputs were reused instead of repeatedly downloading and processing all source data.

11. Ethical and Social Considerations

AI-based disaster prediction is a high-impact application because incorrect predictions can influence how people respond to possible disasters.

False Negatives

If the system predicts a low risk when a disaster actually occurs, users may get a false sense of safety.

False Positives

If the system frequently predicts high risk when there is no disaster, it may create unnecessary fear and reduce people's trust in the system.

Human Responsibility

The system should support human experts and official disaster-management organizations rather than replace them.

Transparency

Users should understand that the predictions are based on historical data and simplified decision-making techniques. Therefore, the results contain uncertainty.

Data Bias

Some areas or time periods may have more complete disaster records than others. This can cause the model to learn from an uneven representation of historical events.

Accessibility

The risk information should be presented using simple terms so that users without technical knowledge can understand the results.

Misuse

The prototype should not be presented as an official disaster early-warning system.

For these reasons, the application clearly informs users that it is an academic prototype and should not be used as a replacement for official disaster warnings.

The project also considers important AI issues such as privacy, bias, accuracy, reliability, misuse, human responsibility and the limitations of automated decision-making.

12. Conclusion

This project developed an AI-based prototype for assessing flood and landslide risk in Sri Lanka by combining Machine Learning, Fuzzy Logic and Rule-Based Reasoning.

Historical climate and disaster data were cleaned and combined with terrain information obtained from elevation data. The final dataset contained **122,736 district-day observations from 24 districts**, using eight environmental features.

Decision Tree and Random Forest models were developed for both flood and landslide prediction. Random Forest was selected as the final model because it provided better overall results.

Validation data were used to select probability thresholds of **0.70 for flood** and **0.80 for landslide**. These thresholds were then kept fixed when testing the models on the unseen test data.

The final Flood Random Forest achieved a test ROC-AUC of **0.915**, while the Landslide Random Forest achieved a test ROC-AUC of **0.9287**. However, the positive-class precision and recall were still relatively low. This was mainly caused by the severe class imbalance and the limited number of recorded disaster events.

Because of these limitations, the project does not depend only on Machine Learning predictions. Instead, the ML probabilities are combined with fuzzy logic and rainfall-based rules to generate easier-to-understand **LOW, MEDIUM and HIGH** risk levels. A Streamlit graphical interface was also developed to allow users to enter environmental conditions and view the resulting flood, landslide and overall risk levels.

Overall, the project demonstrates how several basic AI techniques can be combined to create a practical disaster-risk decision-support prototype.

However, the current system should not be considered an operational early-warning system. More recent data, better spatial information, improved handling of class imbalance, expert validation and real-time weather data would be needed before such a system could be considered for real-world use.

13. References

[1] S. T. Palliyaguru, L. C. Liyanage, O. S. Weerakoon, and G. D. S. P. Wimalaratne, “Random forest as a novel machine learning approach to predict landslide susceptibility in Kalutara District, Sri Lanka,” Proc. 20th Int. Conf. Advances in ICT for Emerging Regions (ICTer 2020), Sri Lanka, 2020, pp. 262–267.

[2] M. A. S. Perera, “A machine learning approach for landslide susceptibility modeling for Rathnapura District, Sri Lanka,” M.S. thesis, Department of Computer Science and Engineering, University of Moratuwa, Sri Lanka, 2021.

[3] N. Hettiarachchi and R. P. Thilakumara, “Water level forecasting and flood warning system: A neuro-fuzzy approach,” International Journal of Applied Engineering Research, vol. 9, pp. 4901–4904, 2014.

14.Contribution Table

INDIVIDUAL CONTRIBUTION SHEET

Student Name: NAH Dilshan

Registration No.: D/DBA/25/0037

Degree / Intake: Data Science and Business Analytics / Intake 42

Group No.: 18

Project Title: AI-Based Flood and Landslide Risk Prediction and Early Warning System

Technical Contributions

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Data Preprocessing & Feature Engineering	Processed climate and disaster datasets, handled missing values, and prepared environmental features for ML training.	Python, Pandas, NumPy	Cleaned dataset, preprocessing code GitHub – Data Preprocessing
Machine Learning Development	Developed and evaluated Decision Tree and Random Forest models for flood and landslide prediction.	Scikit-learn, Machine Learning	Trained models, evaluation results GitHub – ML Development
Model Optimization & Evaluation	Applied balanced class weights, evaluated validation performance, and selected classification thresholds.	Random Forest, Classification Metrics	Confusion matrices, F1, ROC-AUC results GitHub – Model Evaluation
Fuzzy Logic & Risk Integration	Implemented fuzzy membership functions and integrated ML probabilities into the hybrid	Scikit-fuzzy, Python	Fuzzy risk calculations, integrated prediction code GitHub – Hybrid AI Implementation

	risk-assessment process.		
System Integration & Deployment	Contributed to connecting saved Random Forest models with the application's prediction process and deploying the Streamlit system.	Python, Scikit-learn, Joblib, Streamlit,GitHub	Working application, GitHub repository, Models Application source code- GitHub – Streamlit Interface , Live Demo
Streamlit Interface	Contributed to implementing and testing the web-based prediction interface.	Python, Streamlit	Application source code GitHub – Streamlit Interface
Documentation	Contribute to create project documentation with GitHub Readme file	MS Word	GitHub - ReadMe

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  Date: 10/18/2026

INDIVIDUAL CONTRIBUTION SHEET

Student Name: GPN Kaushalya

Registration No.: D/BIS/24/0013

Degree / Intake: Information Systems / Intake 41

Group No.: 18

Project Title: AI-Based Flood and Landslide Risk Prediction and Early Warning System

Technical Contributions

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Data Collection	Assisted in identifying and collecting historical climate and disaster-event datasets for Sri Lanka.	Climate Datasets, DesInventar	Collected datasets, source references GitHub – Data Sources
Data Preparation	Supported dataset inspection, district-name standardization, and checking missing or inconsistent records.	Python, Pandas	Data preparation outputs GitHub – Data Preparation
Feature Engineering	Assisted in preparing rainfall, temperature, and humidity variables for the prediction models.	Python, Pandas, NumPy	Prepared environmental features GitHub – Feature Engineering
Model Testing	Assisted in testing the prediction system using different environmental input conditions.	Python, Streamlit	Test observations, prediction outputs GitHub – PR #1
Documentation	Contributed to preparing project documentation, data descriptions, and presentation materials.	MS Word, PowerPoint	Final report, presentation slides GitHub – Project Documentation

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  Date: 10/17/2026

INDIVIDUAL CONTRIBUTION SHEET

Student Name: AM Weerasinghe

Registration No.: D/BIT/24/0064

Degree / Intake: Information Technology / Intake 41

Group No.: 18

Project Title: AI-Based Flood and Landslide Risk Prediction and Early Warning System

Technical Contributions

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Terrain Data Preparation	Assisted in processing elevation data and preparing terrain-related information for the prediction system.	Python, NASADEM	Terrain processing outputs GitHub – Terrain Processing
Terrain Feature Engineering	Supported the preparation of district-level elevation and slope features.	Rasterio, NumPy, GeoPandas	District terrain dataset GitHub – Terrain Features
Rule-Based Reasoning	Assisted in developing and testing rainfall-based warning rules for risk classification.	Python, IF–ELSE Rules	Rainfall warning logic GitHub – Rainfall Rules
System Testing	Tested the integrated application using different rainfall and environmental conditions.	Python, Streamlit	Test results, screenshots GitHub – System Testing
Technical Documentation	Contributed to documenting system implementation, testing procedures, and project limitations.	MS Word, PowerPoint	Report sections, presentation GitHub – Technical Documentation

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____  __ Date: __17/10/2026_____

INDIVIDUAL CONTRIBUTION SHEET

Student Name: DKNL Jayawardhana

Registration No.: D/BSE/25/0003

Degree / Intake: Software Engineering / Intake 42

Group No.: 18

Project Title: AI-Based Flood and Landslide Risk Prediction and Early Warning System

Technical Contributions

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
System Design	Assisted in designing the workflow connecting ML models, fuzzy logic, rainfall rules, and the user interface.	Python, System Design	System architecture diagram GitHub – System Architecture
Streamlit Interface	Contributed to implementing and testing the web-based prediction interface.	Python, Streamlit	Application source code GitHub – Streamlit Interface
Model Integration	Assisted in connecting saved Random Forest models with the application's prediction process.	Joblib, Scikit-learn	Integrated prediction system GitHub – Model Integration
Functional Testing	Tested user inputs, prediction outputs, and the overall	Python, Streamlit	Prediction screenshots, test outputs GitHub – Functional Testing

	risk-classification functionality.		
Project Documentation	Assisted in preparing technical explanations, implementation details, and final presentation materials.	MS Word, PowerPoint	Final report, presentation GitHub – Project Documentation

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  Date: 10/17/2026